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August 2026

Performance Analysis of Data-oriented Programming in Rust



Bachelor's Thesis

to confer the academic degree of

Bachelor of Science

in the Bachelor's Program

Computer Science

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Abstract

< Abstract in english >

Kurzfassung

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Chapter 1

Introduction

1.1 Motivation

In the past decades, processor speeds have vastly increased due to higher cycle rates and higher core counts, while DRAM has not been able to keep up. This phenomenon is colloquially known as the "memory wall". The resulting problem is that execution speed may no longer be limited by processing power, but by the processor having to wait for data from the DRAM [4]. To address this gap, modern software development has to take into consideration the underlying hardware constraints.

1.2 The Shift to DOP

Object-oriented Programming (OOP) has enjoyed wide adoption and is usually taught long before Data-oriented Programming (DOP) in most curricula. However, OOP poses challenges when it comes to cache-optimization, for example, due to a high degree of coupling and inefficient memory layouts [6]. More modern approaches, such as DOP, aim to mitigate challenges such as these by decoupling logic and data. However, there are more mechanisms that DOP employs to improve cache efficiency and execution time, some of the more significant ones being Single Instruction, Multiple Data (SIMD) vectorization (Section 2.2) and use of the Entity Component System (ECS) software architecture design pattern (Section 2.3).

1.3 Objectives and Methodology

The aim of this thesis is to measure the difference in execution time between OOP and DOP approaches, as the exact difference is hardly ever quantified. The created benchmarks aim to reflect realistic use cases and lean towards the gaming industry, as there has been a significant push in recent years to adopt the ECS pattern. Rust will act as the programming language with the Bevy engine providing the framework. Rust has many characteristics that lend itself well as a benchmarking language [8], such as not having a garbage collector and providing memory safety. The same holds for Bevy and especially the Bevy ECS, as it uses Rust's normal data types instead of requiring more complex features like lifetimes, traits or macros [2]. More intricate details of Rust and Bevy are explained and elaborated on further in section 2.4.

Chapter 2

Background

This section describes the relevant parts of DOP, which also include SIMD vectorization and the ECS design pattern, as they are integral to the nature of the systems under inspection. Furthermore, a brief overview of Rust and the Bevy engine is presented.

2.1 Data-Oriented Programming

The OOP paradigm undoubtedly has many useful features such as encapsulation, polymorphism and inheritance. These features enable software developers to think of problems in terms of objects and their interactions, with objects ideally being self-contained. Due to this, programmers are able to easily reuse and maintain them without having to fear side effects [1]. Nevertheless, objects encapsulating the entirety of their data and logic is a double-edged sword.

As an example, imagine a situation where you may want to update solely the location of some objects, with these objects having further attributes like color, weight etc. Due to encapsulation, the CPU has to load all the data of these objects into its cache, even though it should only have to load the location for its computation. This phenomenon is often called "cache pollution" [4] which poses a problem due to another phenomenon called the "memory wall".

Over the past decades, CPU clock speeds have increased exponentially. So did memory speed, however, at a far slower rate. According to Wulf et al. [12], in 1994 processing speed has been increasing by 80% per year with memory only achieving 8% per year. In the past 30 years, the memory wall has become one of the biggest bottlenecks in high-performance computing, due to the CPU often staying in idle while waiting for data to be fetched from main memory [7].

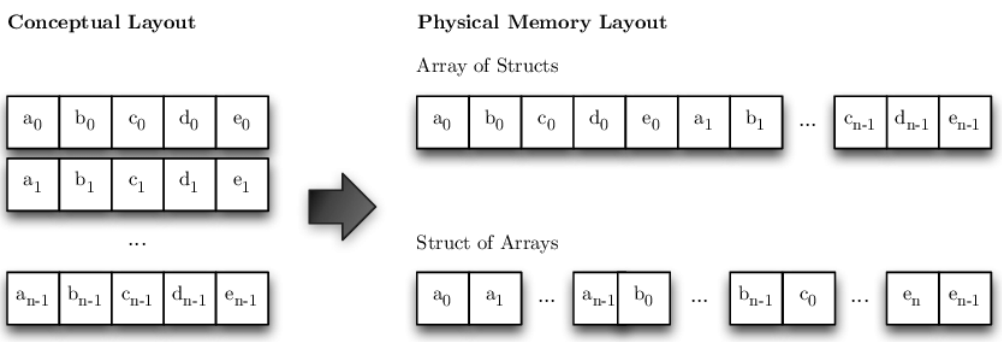


Figure 2.1: Memory Layout Visualization: AoS vs. SoA (taken from [9])

One of the most promising solutions to these problems is DOP, as it aims to decouple data from logic and thus reduce cache pollution and the CPU idling [5]. DOP improves CPU utilization by fundamentally changing how data is handled by moving away from the classical OOP memory layout called Array of Structures (AoS) and instead using a more optimized Structure of Arrays (SoA) layout [5].

In Figure 2.1, both memory layouts are compared. On the conceptual level, there are simply structs with data, e.g. objects with fields. The more conventional AoS layout stores these structs sequentially within an array, thus the name. In contrast, the SoA layout creates multiple arrays which contain only one type of field. As an example, one could want to manage 1000 `Person` objects which possess the fields `String name`, `float weight` and `Color hairColor`. The AoS layout would result in a `Person` Array with 1000 entries. Contrarily, the SoA layout would result in three arrays, one of type `String` containing the `name` values of all 1000 `Person` instances. Similarly, an array is created for `weight` and `hairColor`.

This is due to caches exploiting a principle called spatial locality. Drepper argues [4] that since neighboring data is likely to be used together, it should be loaded into the cache together. This is what the entire idea of SoA is based upon. If one wants to perform a read or write operation on only the `name` of these 1000 `Person` instances, the CPU can load much more relevant data than with the AoS layout, as the cache is not clogged by the irrelevant `weight` and `hairColor` data. Thus, while the SoA layout might appear confusing to programmers used to the AoS layout, it is necessary in order to properly achieve the performance gains granted by SIMD vectorization.

2.2 Data-Level Parallelism via SIMD

Traditionally, to perform operations on data, the CPU has to load these values into a General Purpose Register (GPR). These usually have a fixed size of 32-bit or 64-bit, meaning an instruction can operate on two values of this size at once. For example, assuming a 32-bit environment, a scalar add instruction can add no more than two 32-bit values per cycle.

In order to overcome this limitation, modern CPUs contain specialized SIMD vector units. These units use much wider registers, such as the 512-bit registers found in the Intel MIC architecture, which allow the CPU to treat a single register as a vector of values. By using 512-bit registers, operations can be performed on up to 16 single-precision or 8 double-precision floating-point values in parallel [11]. On consumer grade CPUs, the AVX2 instruction set is widely adopted, using 256-bit registers.

This effectively parallelizes the computation and significantly increases throughput, under the condition that the data is laid out in memory as explained in Section 2.1. When looking at the memory layouts in Figure 2.1 it becomes clear why SIMD vectorization can only bring significant performance gains when the data is stored in a SoA layout. If the data was instead organized as in the AoS layout, SIMD units could not be fed with packed, homogenous data, causing the cache to be filled with irrelevant data that hinders the aspired parallel execution of operations. To achieve the outlined hardware efficiencies without the programmer having to manually manage memory, the ECS design pattern may be employed.

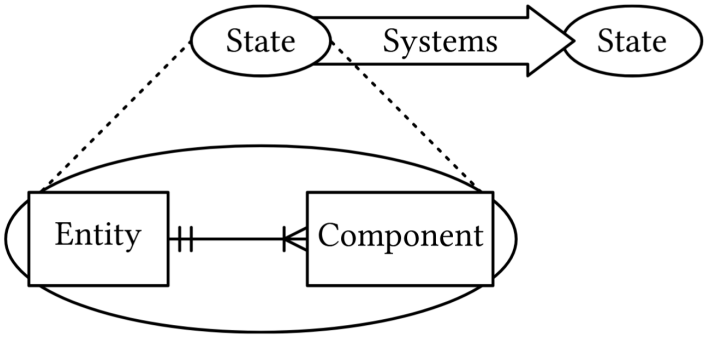


Figure 2.2: Visualization of the ECS pattern depicting (1) Entities, (2) Components and (3) Systems (taken from [10])

2.3 Entity Component Systems

As the presentation by Bilas [3] at the Game Developers Conference in 2002 shows, there has been a desperate need for the ECS pattern since at least the early 2000s. He explains that with increasing complexity of game systems, the resulting hardcoded class tree must undergo repeated refactoring and will naturally stray further and further from clean OOP principles. He noticed that when re-examining the problem of complex object schemes or structures, that he could also view it as a database and that in the end, only the data matters.

This data-centric perspective is the essence of the ECS pattern. It encourages clean separation of data and logic by employing a 3-level architecture as can be seen in Figure 2.2. (1) Entities can be viewed as an identifier value to which components may be attributed. (2) A Component is a domain-specific datatype and its value is comparable to the field value of an object. One entity may possess multiple components. (3) Lastly, a system is a domain-specific piece of logic or behavior. This system performs computations on entities and their components. Entities and Components represent a game state, with the Systems performing the transitions between states [10].

As an example, we revisit the `Person` objects from Section 2.1. We discussed that the SoA memory layout would greatly improve performance, but we did not explain how this layout would be achieved. Traditionally, one would define a `Person` class with fields and methods and then instantiate objects. To transform this into a SoA memory layout, the ECS pattern is applied as follows:

- Entities are created as unique identifiers.
- Components such as `name`, `weight` and `hairColor` are associated with these entities. In an archetype based ECS like Bevy, this ensures that all `weight` components are stored in a contiguous array, separate from `hairColor` [2].
- Lastly, Systems are defined to perform transformations by querying for entities that possess a certain set of components. For example, updating the `hairColor` component of all `Person` instances that possess the `name` and `hairColor` components. This allows the system to iterate over a tightly packed array of only relevant data, enabling the use of SIMD instructions.

2.4 Rust and Bevy engine

Rust is a systems programming language which provides performance comparable to C++ and memory safety through its unique ownership system. The compiler enforces strict rules on how data may be accessed and modified at compile-time. By doing so, Rust eliminates entire classes of bugs such as data races, and without requiring a garbage collector [8].

The Bevy engine expands upon this foundation, using an archetype-based ECS framework that prioritizes performance and modularity. Archetypes group entities with identical combinations of components, which enforces the SoA memory layout [2].

NOTE:
This section specifically will be expanded once the practical part is finished, as explaining Rust and Bevy in detail wouldn't make much sense and would likely be way too long. Focus will be on only the relevant parts for understanding the benchmarks once they are done.

Chapter 3

Benchmarks

3.1 Architecture

- Comparison of architectures (diagrams memory layout)
- Program architecture (what crates, rough program flow)

3.2 Environment

Description of the setup including:

- OS (Ubuntu 26.04)
- Hardware (CPU model/clock speed/core count, RAM, etc.)
- Compiler settings
- Versions of bevy, rapier, criterion etc.
- Execution procedure

Chapter 4

Evaluation

- Benchmark results as diagrams and tables
- Description of the results in text

Chapter 5

Analysis

- Extent of SIMD Vectorization
- Extent of cache pollution
- Comparison of assembly code to showcase differences

Chapter 6

Limitations and Future Work

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List of Acronyms

OOP Object-oriented Programming

DOP Data-oriented Programming

AoS Array of Structures

SoA Structure of Arrays

GPR General Purpose Register

SIMD Single Instruction, Multiple Data

ECS Entity Component System